



NETRA

Decision Intelligence for Disaster Response

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When Disaster Strikes, Every Minute Matters

The Cost of Communication Failure

84% of emergency responses are delayed by communication failures

Manual tower triage takes hours, lives are lost in those minutes

Current approach is reactive, not proactive

84%

response delay risk



Introducing NETRA

Neural Emergency Tower Restoration Algorithm

A decision intelligence engine that optimizes tower restoration for maximum societal benefit

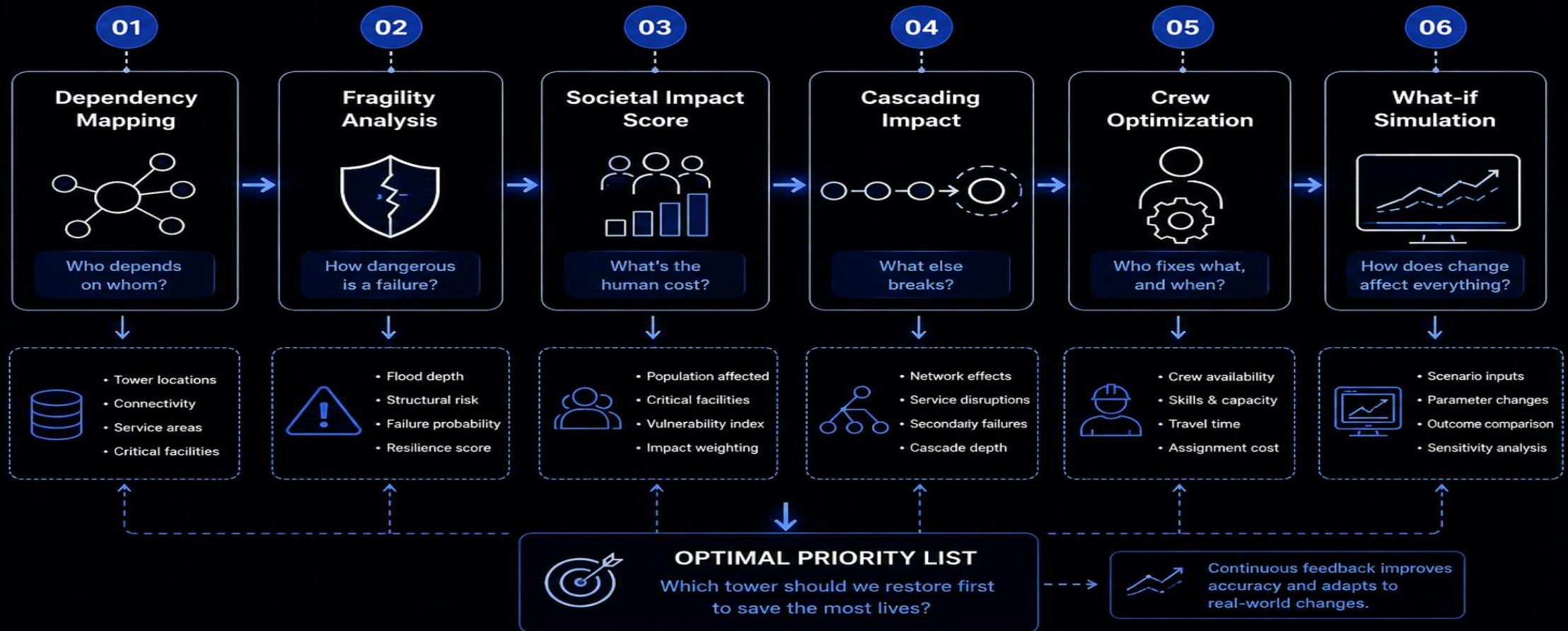


“Which tower should we restore first to save the most lives?”



End-to-End Decision Pipeline

From Data to Action





What Makes NETRA Different

Intelligent. Adaptive. Life-Saving.

Societal Impact Scoring

Weighs population, critical facilities, vulnerability — not just proximity

Cascading Simulation

Models how one failure triggers others

Optimal Crew Dispatch

Hungarian algorithm maximizes societal benefit

Dynamic What-If Analysis

Vary flood levels; priorities shift in real-time

Alternative Comparison

“What if we don’t restore Tower X first?”



Clean, Modular, Production-Ready

Built for Scale

01

Data Ingestion Layer

Collect, validate and unify diverse real-time data



Telemetry
(Tower Status)



Network
(Dependencies)



Population
(Critical Facilities)



Environment
(Flood / Disaster)



Unified
Real-time
Data

02

Decision Engine Core

Analyze, simulate and optimize for maximum societal impact



Impact
Assessment
(Societal Score)



Cascade
Simulation
(Ripple Effects)



Crew
Optimization
(Best Dispatch)



What-If
Analysis
(Scenario Planning)



Optimal
Restoration
Plan

03

Output Layer

Deliver actionable insights for faster, smarter decisions



- ✓ Prioritized restoration plan
- ✓ What-if results
- ✓ Impact breakdown
- ✓ Decision rationale



**Faster decisions.
More lives saved.**



How NETRA Makes Better Decisions

Flood Scenario — 2m water, 5 towers down

TRADITIONAL APPROACH

- Crews sent to closest towers first
- Hospital loses connectivity (no one noticed)
- 18,400 people lose coverage needlessly

NETRA APPROACH

- Identifies T184 as highest impact
- Simulates cascade effects
- 42 min restoration → 18,400 reconnected
- 3 critical facilities restored

Result: Lives Saved. Panic Reduced.



Built for Speed, Ready for Scale

Engine Performance Metrics

< 5 sec

Decision Time

1,000 towers

FULL PIPELINE

What-If Scenarios

per scenario

MOCK → EXASOL

Data Volume

scales with source

EXTERNAL

Configuration

weights & thresholds

Python • NetworkX • SciPy • Pydantic



Meet the Team

One Team. One Mission. Lives First.

Abhishek Yadav

Decision Engine Architect

Abhishek Karthik

Data Pipeline & Integration

Sudeepa Priyadarshini

Impact Modeling & Analytics



Why NETRA Wins

Comparison Table

Criteria	Traditional Dispatch	NETRA
Decision Criteria	Proximity	Societal Impact
Cascading Effects	Ignored	Simulated
What-If Analysis	Manual	Dynamic
Alternative Comparison	None	Built-in
Data Adaptability	Rigid	Any Source
Explainability	Opaque	Clear Reasoning



By the Numbers

Tangible Results from NETRA

25

Towers Analyzed in Bangalore

15,000+

People Reconnected per
Decision

3+

Critical Facilities Restored per
Priority Action

42 min

Restoration Time Optimized

Every second counts. Every life matters.



What's Next

Our Vision for NETRA



v1.0

COMPLETED

Full decision pipeline,
mock data, JSON output



v1.5

IN PROGRESS

Exasol integration, real travel
times



v2.0

FUTURE

ML predictions, real-time sensors,
multi-city



See NETRA in Action

Watch the Decision Engine Live



Visit [https://drive.google.com/file/d/1jC7nBma_LLifbCsII3eGKIzQEtq_Y0l9/view?usp=sharing] to explore the live dashboard



Why This Matters

Human Impact at Scale

Faster response = fewer casualties

Smart resource allocation = lower costs

Transparent decisions = trust in responders

Adaptive planning = resilience in uncertainty



**When disaster strikes, clarity is power.
We bring clarity.**

Join Us in Building Resilient Communities

Let's Build the Future of Disaster Response Together

Partners | Collaborators | Investors

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NETRA — Decision Intelligence for Disaster Response
